

Employee Attrition Analysis and Insight Generation

Data Analytics Report

Executive Summary & Business Insights

This report presents a comprehensive analysis of employee attrition based on the HR analytics dataset. Below are the key data-driven insights derived from statistical methods and exploratory data analysis:

- The overall employee attrition rate is 16.12%.
- Employees working OverTime are 2.9x more likely to leave compared to those who do not.
- The Sales department has the highest attrition rate at 20.63%.
- Employees who travel 'Travel_Frequently' have the highest attrition rate (24.91%).
- Employees who leave earn on average 29.9% less per month (\$4787.09 vs \$6832.74).
- Employees who leave are generally younger (Avg Age: 33.6 years) compared to those who stay (37.6 years).
- Statistically significant numerical drivers of attrition include: Age, DailyRate, DistanceFromHome, EnvironmentSatisfaction, JobInvolvement.
- Statistically significant categorical drivers of attrition include: BusinessTravel, Department, EducationField, JobRole, MaritalStatus.

Machine Learning Validation (Logistic Regression)

A Logistic Regression model was trained to identify the most significant drivers of attrition while maintaining high interpretability.

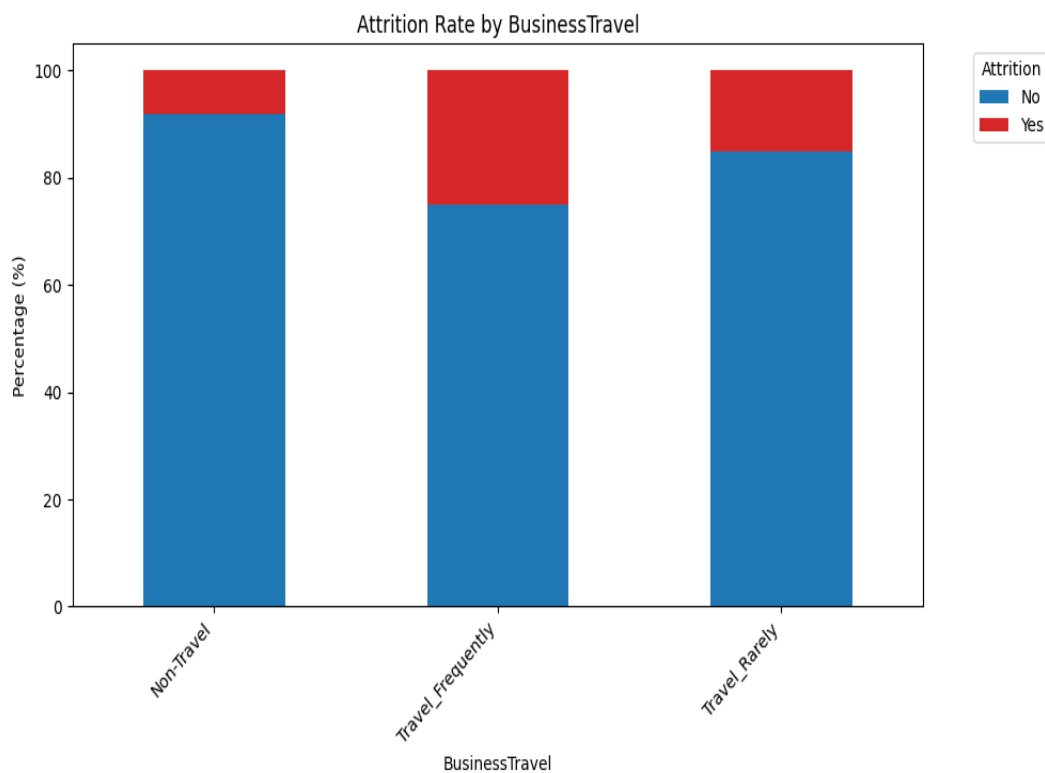
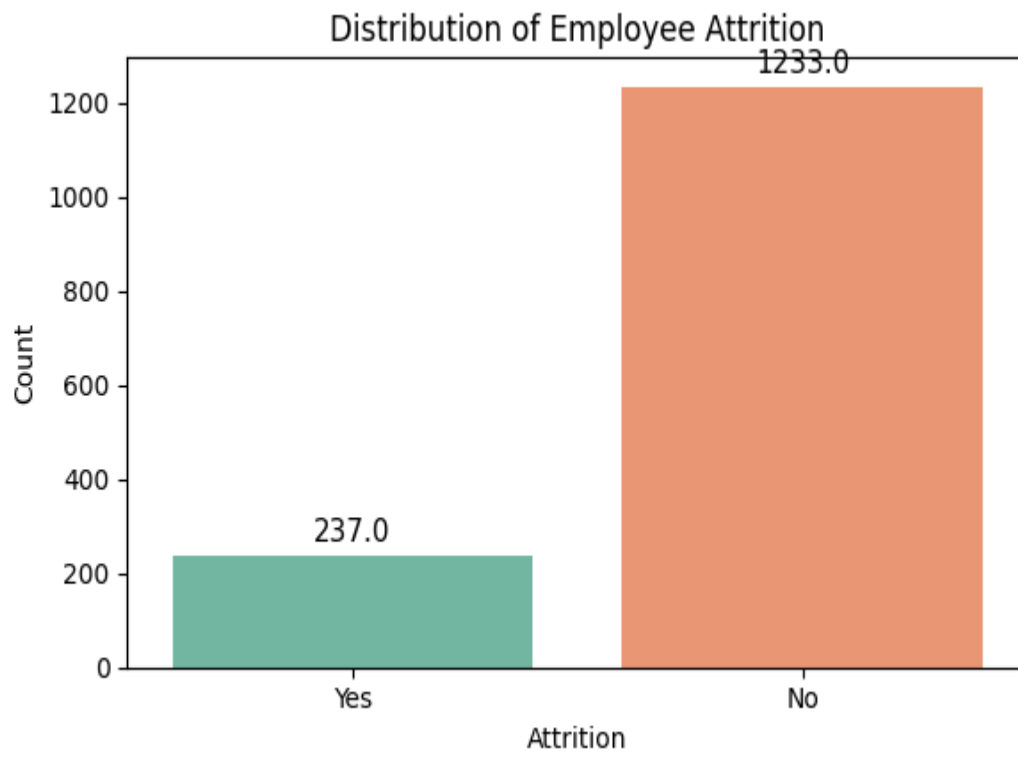
Top Factors INCREASING the probability of Attrition:

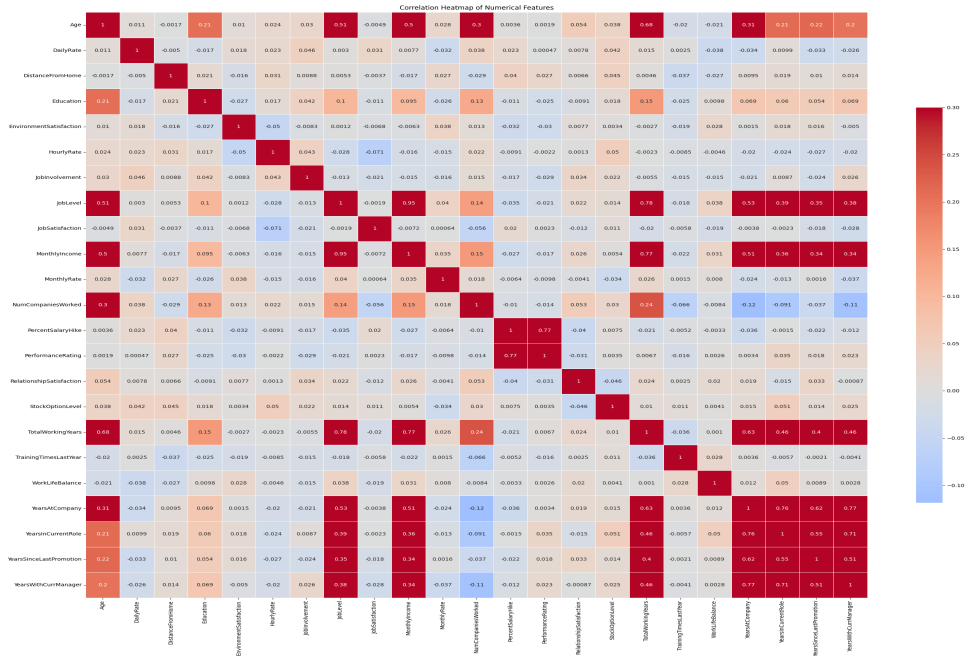
- OverTime: Increases odds of leaving by 5.63x
- BusinessTravel_Travel_Frequently: Increases odds of leaving by 3.58x
- JobRole_Laboratory Technician: Increases odds of leaving by 3.33x
- JobRole_Sales Representative: Increases odds of leaving by 2.00x
- MaritalStatus_Single: Increases odds of leaving by 1.70x

Top Factors DECREASING the probability of Attrition (Retention Drivers):

- Department_Research & Development: Reduces odds of leaving by 72.6%
- EducationField_Other: Reduces odds of leaving by 67.5%
- JobRole_Research Director: Reduces odds of leaving by 59.1%
- EducationField_Medical: Reduces odds of leaving by 52.5%
- EducationField_Life Sciences: Reduces odds of leaving by 42.7%

Key Visualizations





Strategic Recommendations

1. **Compensation Review:** Given that lower income strongly correlates with attrition, HR should benchmark salaries against industry standards, particularly for high-risk roles.
2. **Overtime Management:** High overtime significantly increases the likelihood of leaving. Implement policies to reduce mandatory overtime or provide better compensation/time-off in lieu.
3. **Departmental Interventions:** Departments with the highest attrition should undergo targeted culture and workload assessments.
4. **Career Progression:** Younger employees have a higher attrition rate. Establish clearer career paths and mentorship programs to retain junior talent.